

JNTUK R23 B.Tech AI & DS Curriculum Mapping

Project 17: Smart Attendance & Performance Analyzer

1. Programming & Problem Solving (Python/C Unit)
 - Topic: File Handling (CSV Read/Write), Functions, Data Cleaning
 - Implementation: `load_student_data()`, load/save CSV routines in code/m
2. Linear Algebra & Calculus (Matrices & Vector Spaces Unit)
 - Topic: Feature Matrices, Weight Vector Multiplication, Linear Transfo
 - Implementation: `compute_composite_scores_matrix()` using $S = M \cdot W$
3. Data Structures (Sorting & Searching Unit)
 - Topic: QuickSort Algorithm (Divide-and-Conquer)
 - Implementation: Custom `quicksort_students()` and `_partition()` implemen
4. AI & Machine Learning Foundations
 - Topic: Pearson Correlation, OLS Linear Regression, Rule-Based AI Clas
 - Implementation: `perform_ai_trend_analysis()` & `assign_ai_performance_t`
5. Applied Data Analytics & Visualization
 - Topic: Multi-panel Plotting with Matplotlib & Seaborn
 - Implementation: `plot_performance_dashboard()` saved to `output/demo_scr`